Prtf 2 Danish Monarchs

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Github URL for your rmarkdown/html files: Digital-Methods-HASS/AU788504_Pagh_Mark

Characters: 2,176

1. What function we used to load our dataset in R as **a tibble.** And what we see when we run **head()** on our ‘kings’ digital object:

How do we confirm our kings’ object in R is a tibble?

To confirm our kings’ object in R is a tibble we look at the class output. When you run the R command class() you will see what the data set is made off.

Here is a screenshot of the Console after running **head(kings)** or **glimpse(kings)**:

```
Rows: 57
Columns: 11
$ Name       <chr> "Gorm den Gamle", "Harald 1. Blåtand", "Toke_Gormsen", "Svend 1. Tveskæg", "Harald 2…
$ House      <chr> "Gorm", "Gorm", "Gorm", "Gorm", "Gorm", "Gorm", "Gorm", "Fairhair", "Estridsen", "Es…
$ Start_year <dbl> 936, NA, 985, NA, 1014, 1018, 1035, 1042, 1047, 1074, 1080, 1086, 1095, 1104, 1134, …
$ End_year   <dbl> 958, NA, 986, NA, 1018, 1035, 1042, 1047, 1074, 1080, 1086, 1095, 1103, 1134, 1137, …
$ Birth_year <dbl> 908, 936, NA, NA, 963, NA, 995, 1018, 1024, 1019, 1041, 1043, 1050, 1055, 1065, 1100…
$ Death_year <dbl> 958, 985, 986, 1014, NA, 1035, 1042, 1047, 1076, 1080, 1086, 1095, 1103, 1134, 1137,…
$ Gender     <chr> "M", "M", "M", "M", "M", "M", "M", "M", "M", "M", "M", "M", "M", "M", "M", "M", "M",…
$ Dynasty    <chr> "Jellingdynastiet", "Jellingdynastiet", "Jellingdynastiet", "Jellingdynastiet", "Jel…
$ Source     <chr> "https://da.wikipedia.org/wiki/Gorm_den_Gamle", "https://da.wikipedia.org/wiki/Haral…
$ BirthDMY   <chr> "908", "936", NA, "17/04/963", "994", "995", "1018", "1024", "1019", "1041", "1043",…
$ DeathDMY   <chr> "958", "987", NA, "03/02/1014", "1018", "12/11/1035", "08/06/1042", "25/10/1047", "2…
>
```

A tibble: 6 x 11

Name <chr>	House <chr>	Start_year <dbl>	End_year <dbl>	Birth_year <dbl>	Death_year <dbl>	Gender <chr>	Dynasty <chr>
Gorm den Gamle	Gorm	936	958	908	958	M	Jellingdynastiet
Harald 1. Blåtand	Gorm	NA	NA	936	985	M	Jellingdynastiet
Toke_Gormsen	Gorm	985	986	NA	986	M	Jellingdynastiet
Svend 1. Tveskæg	Gorm	NA	NA	NA	1014	M	Jellingdynastiet
Harald 2.	Gorm	1014	1018	963	NA	M	Jellingdynastiet
Knud 1. den Store	Gorm	1018	1035	NA	1035	M	Jellingdynastiet

6 rows | 1-8 of 11 columns

When we run ‘head()’ on our ‘kings’ digital object we see 11 columns and 57 rows, which is the total amount of data in the CSV file. When we run glimpse, we see the first 6 rows of data.

2. The average age of reign among the Danish monarchs and how many rulers that have exceeded it:

To find the average age of reign among our danish monarchs we use this code in R:

Code:
```
kings %>%
  select(Start_year, End_year) %>%
  mutate(duration = End_year - Start_year) %>%
  summarise(duration = mean(duration, na.rm = TRUE))
```

A tibble: 1 × 1

duration <dbl>
19.57692

1 row

Result: **The average age of reign is 19.57692 years.**

Then we want to find out how many of the danish rulers sat on the throne for longer than the average age of reign of 19.57692 years, we use this code in R:

```
kings %>%
  select(Name, Start_year, End_year) %>%
  mutate(duration = End_year - Start_year) %>%
```

```
filter(duration > mean(duration, na.rm = TRUE))
```

Name <chr>	Start_year <dbl>	End_year <dbl>	duration <dbl>
Gorm den Gamle	936	958	22
Svend 2. Estridsen	1047	1074	27
Niels	1104	1134	30
Valdemar 1. den Store	1157	1182	25
Knud 4.	1182	1202	20
Valdemar 2. Sejr	1202	1241	39
Erik 5. Klipping	1259	1286	27
Erik 6. Menved	1286	1319	33
Valdemar 4. Atterdag	1340	1375	35
Erik 7. af Pommern	1396	1439	43
Chrstian 1.	1448	1481	33
Hans	1482	1513	31
Christian 3.	1536	1559	23
Frederik 2.	1559	1588	29
Christian 4.	1588	1648	60
Frederik 3.	1648	1670	22
Christian 5.	1670	1699	29
Frederik 4.	1699	1730	31
Frederik 5.	1746	1766	20
Christian 7.	1766	1808	42
Frederik 6.	1808	1839	31
Christian 9.	1863	1906	43
Christian 10.	1912	1947	35
Frederik 9.	1947	1972	25
Margrete 2.	1972	2023	51

So out of the 54 danish rulers 25 of the rulers were on the throne for more than 19.57692 years. We see the 25 rulers above, and they have a period of reign between 20-60 years. Christian 4. ruled the longest time for 60 years.

3. The number of days the three longest-ruling monarchs ruled:

To find the three longest-ruling monarchs, we use this code in R:

```
Code: kings %>%

  select(Name, Start_year, End_year) %>%

  mutate(duration = End_year - Start_year) %>%

  arrange(desc(duration)) %>%

  slice(1:3) %>%
```

```
mutate(Days = duration * 365)
```

A tibble: 3 x 5

Name <chr>	Start_year <dbl>	End_year <dbl>	duration <dbl>	Days <dbl>
Christian 4.	1588	1648	60	21900
Margrete 2.	1972	2023	51	18615
Erik 7. af Pommern	1396	1439	43	15695

3 rows

Result:

The three longest-ruling monarchs are Christian 4., Magrete 2. and Erik 7. af Pommern. Christian 4. ruled for 21.900 days, Margrete 2. ruled for 18.615 days and Erik 7. af Pommern ruled for 15.695 days.